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$$\begin{aligned}f(x) &= \frac{x^2 - 2 - 2}{x^2 + 5x + 6} \\f'(x) &= \frac{(x^2 + 5x + 6)(2x - 1) - (2x + 5)(x^2 - x - 2)}{(x^2 + 5x + 6)^2} \\&= \frac{2x^3 + 9x^2 + 7x - 6 - 2x^3 - 3x^2 + 9x + 10}{x^4 + 25x^2 + 36 + 10x^3 + 12x^2 + 60x} \\&= \frac{6x^2 + 16x + 4}{x^4 + 10x^3 + 37x^2 + 60x + 36} \\&= \frac{2(3x^2 + 8x + 2)}{(x + 2)^2(x + 3)^2}\end{aligned}$$

References